

Convexity

Def. f convex $\iff$ epi(f) convex $\iff f(\theta x + (1 - \theta)y) \leq \theta f(x) + (1 - \theta)f(y) \ \forall \theta \in [0, 1]$.

Characterizations (use these):

- Twice diff.: $\nabla^2 f(x) \succeq 0 \ \forall x$ (evals for $x^\top Qx$).
- 1st-order: $f(y) \geq f(x) + \nabla f(x)^\top (y - x) \ \forall x, y$.
- Monotone grad: $(\nabla f(x) - \nabla f(y))^\top (x - y) \geq 0$.
- $f^{**} = f$ (proper, closed epi).
- Restriction to any line is convex.

Preserving ops: nonnegative lin. comb., pointwise **max**, affine comp. $f(Ax + b)$, intersection of convex sets.

Does *NOT* imply convex (traps + counterex.):

- Unique/global or single local min ($f(x) = |x|$ on $[-1, 1]$, 1 outside).
- $\nabla^2 f$ bounded below (must be $\succeq 0$).
- $|\nabla f(x) - \nabla f(y)| \geq \mu |x - y|$ ($\mu > 0$).
- Pointwise **min** of convex: e.g. $f_1 = (x + 1)^2, f_2 = (x - 1)^2$.
- Product $f_1 f_2$: e.g. $x \cdot (-x) = -x^2$ (concave).
- $\exp(f(x))$ or $g(x) = 1/\ln x$ ($x > 1$) (Hessian neg.).
- $\sqrt{x}$ on $[0, 1]$: no finite Lip. const. ($\nabla f \rightarrow \infty$ at 0).

Examples:

- $\frac{1}{2}x^2 + p \ln x$ on $[1, \infty)$: convex $\iff p \leq 1$ ($\nabla^2 = 1 - p/x^2 \succeq 0$).
- $1/x_i$ convex on $\{x > 0\}$.
- Quadratic $x^\top Qx$ convex $\iff Q \succeq 0$.

Tactic (MC/TF): Test Hessian/1st-order or give counterex. Local min = global $\nRightarrow$ convex.

Conjugate $f^*(y) = \sup_x (y^\top x - f(x))$

f proper convex closed:
 $f^{**} = f. \ y \in \partial f(x) \iff x \in \partial f^*(y).$

4-Step Recipe:

- Set $\nabla_x (y^\top x - f(x)) = 0 \implies \nabla f(x) = y$.
- Solve for stationary x_y .
- If $x_y \in \text{int}(\text{dom } f)$, $f^*(y) = y^\top x_y - f(x_y)$.
- Restricted domain $[a, b]$: evaluate boundaries ($x_y < a \rightarrow x = a, x_y > b \rightarrow x = b$).

Quiz Examples:

- $f(x) = -x^2, x \in [-1, 1]$: $f^*(y) = 1 + |y|$ (V-shape plot, label axes).
- $f(x) = x^2 + x, x \in [-1, 1]$: stationary $x = \frac{1}{2}(y - 1)$,

$$f^*(y) = \begin{cases} -y & y \leq -1 \\ \frac{1}{4}(y - 1)^2 & -1 \leq y \leq 3 \\ y - 2 & y \geq 3 \end{cases}$$

- $f(x) = -\ln x$ ($x > 0$): $f^*(y) = -1 - \ln(-y)$ ($y < 0$), ∞ else.
- $f(x) = e^x$: $f^*(y) = y \ln y - y$ ($y > 0$), ∞ else.
- $f(x) = \frac{1}{2}x^\top Qx$: $f^*(y) = \frac{1}{2}y^\top Q^{-1}y$.

Tactic: Plot if asked (label axes). Stationary often quadratic; always check boundaries.

Subdifferential & Optimality

$$\partial f(\bar{x}) = \{v \mid f(x) \geq f(\bar{x}) + v^\top (x - \bar{x}) \ \forall x\}.$$

Optimality: $0 \in \partial f(x^*) \iff x^*$ global minimizer (unique if strict convex).

Example $f(x) = |x|$:

$$\partial f(x) = \begin{cases} \{-1\} & x < 0 \\ [-1, 1] & x = 0 \\ \{1\} & x > 0 \end{cases}$$

Minimizer at points where subdifferential plot contains 0 (vertical segment over $y = 0$).

Tactic (plot Q): Find point where subdiff. interval/segment includes 0.

Projections (PGD): $x_{k+1} = \text{prox}_{T,C}(x_k - T\nabla f(x_k))$.

- $\|x\|_\infty \leq 1$: clamp each coord to $[-1, 1]$.
- $\|x\|_2 \leq 1$: $z/\max(1, \|z\|_2)$.
- ℓ_1 -ball: soft-thresholding.

Duality, Lagrangian, KKT

Lagrange Duality

Consider $f^* = \inf_{x \in \mathbb{R}^n} f(x)$ s.t. $g(x) \leq 0, h(x) = 0$ (1)

Lagrangian $\mathcal{L}(x, \lambda, \nu) = f(x) + \lambda^\top g(x) + \nu^\top h(x)$

Dual Function $d(\lambda, \nu) = \inf_{x \in \mathbb{R}^n} \mathcal{L}(x, \lambda, \nu)$

Weak duality: $d^* \leq p^*$ always.

Strong duality + KKT (convex + Slater: $\exists \tilde{x}, g_i(\tilde{x}) < 0$):

$$\begin{aligned} \nabla_x \mathcal{L}(x^*, \lambda^*, \nu^*) &= 0, \\ g(x^*) &\leq 0, \ h(x^*) = 0, \ \lambda^* \geq 0, \\ \lambda^{*\top} g(x^*) &= 0 \text{ (compl. slackness)}. \end{aligned}$$

KKT Conditions (Karush-Kuhn-Tucker)

If Slater's condition holds and OP is convex $\rightarrow x^* \in \mathbb{R}^n$ is a minimizer of the primal and $(\lambda^* \geq 0, \nu^*) \in \mathbb{R}^{n_g} \times \mathbb{R}^{n_h}$ is a maximizer of the dual $\Leftrightarrow$ to the following statements:

KKT-1 (Stationary Lagrangian) $\nabla_x \mathcal{L}(x^*, \lambda^*, \nu^*) = 0$

KKT-2 (primal feasibility) $g(x^*) \leq 0, h(x^*) = 0$

KKT-3 (dual feasibility) $\lambda^*, \nu^* \in \mathbb{R}^{n_h} \geq 0$

KKT-4 (complementary slackness) $\lambda^{*\top} g(x^*) = 0, \nu^{*\top} h(x^*) = 0$

In addition we have: $\sup_{\lambda \geq 0, \nu \in \mathbb{R}^{n_h}} q(\lambda, \nu) = \inf_{x \in C} f(x)$

Example $\min \frac{1}{2}\|a\|^2$ s.t. $y_i(a^\top x_i + b) \geq 1$; dual $\max_{\lambda \geq 0} d(\lambda)$. Solves to $\lambda_A^* = \lambda_B^* = 1/4$.

Example $f = 1/x_1 + 4/x_2, x_1 + x_2 \leq 3, x_{1,2} \geq 1$: write full $\mathcal{L}$, solve stationarity+slack.

Tactic: Write full $\mathcal{L}$, set $\nabla_x \mathcal{L} = 0$, substitute x^* into $d(\lambda)$, maximize (often quadratic). Check Slater. Compl. slackness key.

Problem Classes (Hierarchy)

$$\text{LP} \subset \text{QP} \ (P \succeq 0) \subset \text{QCQP} \subset \text{SOCP} \subset \text{SDP}.$$

2024 Trap: $\min f^\top x$ s.t. $|A_i x + b_i|^2 \leq c_i^\top x + d_i, Fx = g$:

- SOCP: yes ($\|A_i x + b_i\|_2 \leq c_i^\top x + d_i$).
- Not** always QP/QCQP/SDP (quadratic-over-linear constr. not quadratic form).

Smoothness, Strong Conv., Rates

$$\mu\text{-str. conv.} + L\text{-smooth} \implies \kappa = L/\mu.$$

Defs:

- L -smooth: $\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\| \iff \nabla^2 \preceq LI$.
- μ -str. conv.: $f(y) \geq f(x) + \nabla f(x)^\top (y - x) + \frac{\mu}{2}\|x - y\|^2 \iff \nabla^2 \succeq \mu I$.
- Lip. convex (L_f): subgradient method.

Assumptions	Method	$f(x_N) - f(x^*) \leq \varepsilon$
μ -sc + L -sm	GD/PGD	$N \sim \kappa \ln(1/\varepsilon)$
μ -sc + L -sm	Nesterov	$N \sim \sqrt{\kappa} \ln(1/\varepsilon)$
L -sm (no μ)	Nesterov	$N \sim \sqrt{LR^2/\varepsilon} \ (O(1/\sqrt{\varepsilon}))$
Lip. convex (L_f)	Subgrad.	$N \sim (RL_f/\varepsilon)^2 \ (O(1/\varepsilon^2))$

Tactic: Match assumptions exactly. Projected GD has same rate as GD (cheap proj.: simplex, ℓ_1/ℓ_∞ -balls). $\sqrt{x}$ on $[0, 1]$: no Lip. const.

Handy Formulas & Tactics

Matrix: $\nabla(x^\top Ax) = (A + A^\top)x, \nabla^2 = A + A^\top$.

Quadratic $f(x) = \frac{1}{2}x^\top Qx + b^\top x$: min at $-Q^{-1}b$ ($Q \succ 0$), $\mu = \lambda_{\min}(Q), L = \lambda_{\max}(Q)$.

2x2 Eigenvalues: $\lambda^2 - \text{Tr}(A)\lambda + \det(A) = 0 \implies \lambda = \frac{\text{Tr} \pm \sqrt{\text{Tr}^2 - 4 \det}}{2}$.

$$\text{2x2 Inverse: } A^{-1} = \frac{1}{\det(A)} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

Schur (KKT):

$$\begin{bmatrix} Q & A^\top \\ A & 0 \end{bmatrix} \begin{bmatrix} x \\ \nu \end{bmatrix} = \begin{bmatrix} -c \\ b \end{bmatrix} \implies \nu = -(AQ^{-1}A^\top)^{-1}(AQ^{-1}c+b), \ x = -Q^{-1}(A^\top \nu + c).$$

General Tips:

- Weak duality always holds. Strong duality + KKT under convexity + Slater.
- $\partial f(0) = [-1, 1]$ for $|x|$.
- Check domains ($x > 0$ for $\ln$).
- Common conjugates: $\pm x^2$ (restricted), $\ln, \exp, 1/x_i, x^2 + x$, norms.
- One A4 priority: convexity traps, conjugate recipe+examples, KKT steps, rates table, matrix hacks.